Supplementary reproducibility and verification record

22 October 2026

1 Purpose and mathematical scope

Revision 7 configures the author-supplied repository address and planned release links without changing the scientific result. It retains the complete execution and connection audit in Section 7 and documents the restricted infinite input class in Section 5. Earlier records are explicitly dated. The frozen scientific sources and fixed-instance numerical barrier are unchanged.

This record links the inputs, executions and arithmetic checks to the analytic reductions in the manuscript and the stated arithmetic assumptions.

The unperturbed result concerns one neutral dipole, with charges $+1$ and -1 on plaquettes centered at $(0,0)$ and $(12,0)$, respectively, for the exact cosine interaction on the infinite square lattice. Each unoriented nearest-neighbor bond is counted once:

$$E(\theta) = \sum_{e} \bigl(1 - \cos D_e\theta\bigr), \qquad \|\theta\|_{\mathcal{X}} = \|D\theta\|_{\ell^2}.$$

Here $\mathcal{X}$ is the homogeneous energy completion modulo constants, with the conventions specified in the manuscript. The target set is the set of all physical uniform lifts,

$$\mathcal{K} = \{u \in \mathcal{X} : D_e u \in 2\pi\mathbb{Z} \text{ for every } e\}.$$

The minimax ranges over all strong-$\mathcal{X}$ continuous paths from the certified minimum to $\mathcal{K}$. No reflection constraint is imposed on the paths or their transverse perturbations, and the infinite exterior is included.

The certificate establishes, for this configuration,

$$B_{\mathrm{ann}} = E(s_*) - E(m_*), \qquad 0.00567867 < B_{\mathrm{ann}} < 0.00570116,$$

where s_* is either member of the reflected asymmetric index-one saddle pair. The separating construction uses 109 charts and 436 coordinate faces, with exact-saddle caps and full cutting-plane checks. These are finite verification objects controlling the full-space problem; they do not replace it by a finite-box variational problem.

Meaning of reproduction. The run documented here used a new extraction and new output directory, recompiled the original checking sources, and reran the default supervisor. It reused fixed candidate arrays and proposals. It did not use archived result files as mathematical premises or resume cached certificates. This is a fresh reproduction with the same checking software, not a second, independently implemented interval engine. It is also not a proof-assistant formalization or external journal peer review.

2 Frozen archives and integrity

All three original source archives are required. The optimality extension deliberately depends on the separator and connection packages rather than duplicating their large proposal arrays. Extract them into one parent directory, retaining their original directory names:

Package	Extracted directory
Separator	`xy_global_separator`
Connection	`xy_global_connection`
Optimality extension	`xy_optimality_extension`

The following sizes are exact byte counts, not rounded display sizes. The SHA-256 values identify the archives used in the reported run.

S2: separator source archive

`XY_Separacion_Global_Reproducible_2026-10-19.zip`
Size: 228 006 035 bytes. SHA-256:

`47eaf4886bc10b7bbb7ef259c44bfef9134bf3402920594ca6e20cd87235cedc`

This supplies the original finite atlas, boundary ledger, proposals, checking sources, and associated analytic records for global separation.

S3: connection source archive

`XY_Conexion_Global_Reproducible_2026-10-19.zip`
Size: 441 730 856 bytes. SHA-256:

`3a4163d8b24c21f95acbdf502ceaed46e967180c21abb363d3436ae5e2afb042`

This supplies the stationary-state prerequisites, infinite-tail input, upper-path construction, and the original connection checking sources.

S4: optimality-extension source archive

`XY_Optimalidad_Extension_Reproducible_2026-10-19.zip`
Size: 59 642 630 bytes. SHA-256:

`42034c997678fa511ff261667ccb18edceee6d6bb05e543ddb6604f82958615d`

This supplies `verify_optimality.py`, exact-saddle cap and adjoint checks, cutting-plane membership checks, and the optimality assembly.

Integrity results. The reproduction checked 1 808 original archive members and 548 fresh-output hashes. The integrity status was `PASS`, with no mismatches. Original members remained unmodified. The manuscript/checker archive S1 contains the present source, reproducible conceptual-figure code and vector assets, audit summaries and arithmetic checkers; its own final archive hash belongs in the accompanying package manifest, avoiding a self-referential hash inside that archive.

3 Historical fresh execution, 20 October 2026

Run the following command from `xy_optimality_extension`, with an initially empty or absent `results_fresh` directory. The line breaks below are shell continuations of the exact recorded command:

```
python3 verify_optimality.py \
  --separator-package ../xy_global_separator \
  --connection-package ../xy_global_connection \
  --output results_fresh --workers 2
```

Do not supply the supervisor's validated-results or cached-results options when reproducing this fresh mode. Python's standard library and a suitable C++17 compiler suffice for the certificate; numerical candidate search software is not needed to rerun the checking chain.

The observed run started at `2026-10-20T11:16:33.720428+00:00` and ended at `2026-10-20T11:29:17.084643+00:00`. Its wall time was 763.3642072677612 seconds. The recorded environment was Python 3.12.14, g++ 13.3.0, Linux x86-64 with glibc 2.39, and little-endian storage. The compiler flags were

```
-std=c++17 -O2 -DNDEBUG -fno-fast-math
-ffp-contract=off -frounding-math
```

The source guards accept IEEE binary64 and `longdouble` with at least 64 mantissa bits. The audited reproduction target is narrower: exactly 64 significand bits, little-endian input and round-to-nearest. The companion preflight enforces that target. Wider arithmetic requires a separate serialization analysis; successful compilation is insufficient.

Memory observations. The single-child peak `ru_maxrss` was 1 874 192 KiB. The largest periodically sampled aggregate descendant resident set was 2 476 680 KiB. Sampling can miss a larger transient aggregate peak. These are observations from this execution, not portable resource bounds.

Expected terminal record. The new `results_fresh/certificate.json` had status `FRESH_OPTIMAL_ANNIHILATION_BARRIER_PASS` and SHA-256

```
2bcb4121d588ecef7a2f8e8ad9036321d1286bf90d39d4aaa7d9713d45a5d763
```

The nested phases reported `FRESH_GLOBAL_SEPARATOR_PASS`, `INTEGRAL_MINIMUM_SADDLE_SUBLEVEL_PASS`, and `FRESH_GLOBAL_VARIATIONAL_SADDLE_CONNECTION_PASS`. Their reported durations were 182.741558, 406.065284, and 490.923117 seconds, respectively; these phases are nested and must not be added as disjoint tasks. The default supervisor recorded no skipped checks. Its summaries and captured standard-output/error logs accompany S1.

4 New arithmetic checks and analytic obligations

The two new Python checkers use exact rational arithmetic through `fractions.Fraction`. They check consequences of analytic arguments; they are not replacements for those arguments or for the original interval engines. Their source headers and JSON records make this distinction explicit.

An additional payload checker, `verify_path_inputs.py`, inspects the original binary arrays, zero boundary values, manifests, and elementary norm estimates used to obtain $M = 6007$. It reproduces a previously developed input audit, rather than adding another interval engine. Its source and new output record accompany S1. The correction and geometry enclosures it consumes remain dependencies of the separately reproduced global chain.

Coupling robustness

`verify_robustness_constants.py` reads four fresh records: the global certificate, the minimum and saddle prerequisite certificates, and `caps.json`. It checks their statuses, relevant exact input bounds, and records their hashes. For $J_e = 1 + b_e$, $\|b\|_{\ell^1} \le 1/500$, its rational tests verify the continuation and barrier constants used in the manuscript. In particular, with $\sigma = 57/1000$, $\tau = 71/100$ and $r = 1/32$,

$$\sigma r - \tau/500 - \tau r^2/2 = \frac{373}{25600000} > 0, \quad \kappa = \frac{371}{912} < 1, \quad \gamma_b = \frac{541}{16000} > 0.$$

The resulting record has status `COUPLING_ROBUSTNESS_RATIONAL_CHECKS_PASS`. It also checks the positive charge margin, separation of the continued saddle branches, the low-energy bridge, and constants for the specified $\mathcal{X}$-Riesz gradient flow. The analytic bounded-oscillation comparison gives

$$|B_b - B_0| \le 2\|b\|_{\ell^1}, \qquad 0.00167867 < B_b < 0.00970116.$$

The checker neither solves new perturbed equilibria numerically nor reexecutes the old global chain. It does not show that a perturbed minimax is attained at a continued saddle, or that the dominant mechanism changes.

Green-tail localization budget

`verify_green_tail_budget.py` checks explicit rational consequences of the separately proved gradient-projector kernel and tail estimates. For the original certified path inputs it uses $M = 6007$, $R_* = 4097$, $a = 27/500000$, and safe rational majorants for $32\sqrt{2}$ and $\sqrt{227}$. The critical-point recurrence is

$$T(R) \le \frac{32\sqrt{2}Q}{R} + aT(R/2), \qquad 2a < 1.$$

The stated A/R estimate is at the specified dyadic radii; an arbitrary intermediate radius uses the additional factor 2. The checker reports `EXACT_GREEN_TAIL_BUDGET_ARITHMETIC_PASS`. For allocated localization error $\varepsilon/4$, it obtains

ε	n	$R_* 2^n$
10^{-2}	54	73804990693347688448
10^{-6}	68	1209220967519808527532032

No box of either size was materialized. These conservative bounds make the localization effective mathematically; they do not make a full-grid computation at those radii practical and do not give a new numerical barrier enclosure. The analytic kernel proof remains in the main text.

5 Verification boundaries and reuse

The fresh supervisor includes both reflected projected normals and their exterior tails; distances to all 109 chart centers; two adjoint equations at size 1024 compared with Neumann size 2048; the five additionally enlarged charts $42, 43, 44, 53, 55$; exact-saddle caps; retained boundary walls and both entire exact cutting planes; minimum membership; exclusion of every physical uniform lift; and the final rational identity and strict barrier bracket. This list explains the global status; an interior stationary residual or a finite-box Hessian alone would not suffice.

The fresh default run does not include candidate search, historical attack scripts outside the supervisor, a separately implemented secondary boundary checker, or the separate heteroclinic extension. The present reproducibility claim therefore does not establish a heteroclinic orbit, a thermal escape rate, universal XY metastability, or optimality for arbitrary configurations and separations. Internal mathematical cross-checks of the new analytic deductions are not an independent implementation of the original global certificate and are not external peer review.

Targeted source checks, 20 October 2026

The additional inspection recorded under `verification_notes/current_targeted` has a narrower scope than the historical fresh global run. The optimality archive matched all 593 members in its file manifest, and the original manuscript archive matched all 48 entries in its manifest. The subsidiary robustness and Green arithmetic outputs were regenerated and matched their archived values. Hash tampering and a missing saddle record were rejected by the subsidiary robustness checker; these are not tests of every failure mode of the global supervisor.

The exact-arithmetic boundary and membership programs were also rerun using the archived local-oracle outputs: 109 charts and 2208 coordinate-face evaluations left no unresolved pieces. The resulting status is explicitly conditional. The minimum radial and circle energy margins above the supplied comparison level were respectively greater than $6.79\,10^{-6}$ and $9.61\,10^{-7}$; the largest accepted fiber-minimizer distance was less than $0.163746 < 0.165$. This replay checks the geometric assembly, not the construction of its local-oracle, residual, or exterior inputs from the antecedent raw arrays unavailable during this additional inspection. The two antecedent archives are still required for a new full-chain execution.

The inspected C++ backend uses its own trigonometric and inverse-tangent series with explicit remainder bounds. A finite set of 1166 diagnostic comparisons against separately evaluated high-precision formulas found no discrepancies; such tests do not prove an interval contract for all inputs. The consumed endpoints of the four extension modules use an additional outward step before 30-digit decimal serialization. The supported target is therefore restricted here to a 64-bit `longdouble` significand, binary64 little-endian input, and round-to-nearest. The new `verify_supported_environment.py` checks that target; it does not execute or validate the global certificate. Wider arithmetic formats are not justified by a guard that merely accepts at least 64 significand bits.

Two general implementation limitations remain documented in the source inspection report. An injected NaN is not rejected by the internal congruence helper, although nonfinite raw input is rejected and no such intermediate was shown for the frozen data. The historical standalone `run()` decimal output also lacks a general outward-serialization guarantee; the four inspected extension modules rename that main and do not invoke it. At this inspection stage these observations neither established a counterexample to the fixed certificate nor audited its antecedent implementations; Sections 6 and 7 record the later work. The immutable originals were not altered. Hardening a frozen engine would require updating its pins and repeating every dependent verification.

The revised Green-budget checker replaces Python assertions by explicit checks that remain active under `python -O`; normal and optimized runs agree, and nonpositive tolerances are rejected. The cap arithmetic also has the transparent rational lower margin $k_c \geq 19/40000$, given the certified transverse floor 1/25 and radius 33/200.

Reusable a posteriori conditions. To apply the method to another configuration, supply finite-energy candidate representations with certified full-space residuals, branch and exterior remainders; the finite comparison matrices and their interval inequalities; the complete boundary ledger; an upper path with analytic exterior control; and every endpoint/membership check needed by the minimax argument. The abstract sufficient conditions in the manuscript describe these obligations. The supplied program is not an unconditional algorithm guaranteed to certify every configuration. Failure of a sufficient test proves neither instability nor a cheaper escape.

What a subsequent verification should compare. First identify the three input archives by byte count and SHA-256. Then run the fresh command in a separate output directory, inspect every phase status and skipped-check field, and retain the produced logs and manifest. Compare mathematical bounds and final statuses as well as hashes: build metadata or other harmless output formatting can differ on a new system. A genuinely independent implementation would additionally reconstruct the analytic-to-computational reductions and interval operations with a separate checking codebase. That stronger claim is not made here.

Supplement inventory and availability. The accompanying `INDEX.txt` and `INDEX.html` describe five items: S1, the small source and checker snapshot; S2–S4, the three frozen source archives; and S5, this readable record. The repository is `https://github.com/rsumoza/global_annihilation_barriers_infinite_-XY_lattice`. The planned frozen release is `siads-v7-2026-10-22`, at `https://github.com/rsumoza/global_annihilation_barriers_infinite_-XY_lattice/releases/tag/siads-v7-2026-10-22`. Publication and remote downloads have not yet been verified. The index contains the intended asset URLs, byte counts, hashes and exact replay commands. S1 alone cannot reproduce the global chain. The source commit is resolved from the published tag and recorded externally in the release description and journal submission record. No DOI is asserted. The index and this PDF accompany the initial submission; any rejected archive size must be addressed through an accepted editorial transfer route.

New infinite input class. The manuscript's section on a terminating local input class proves reuse of the audited communication height for block-generated fields within energy distance 1/20 of the certified minimum. The inputs can have infinitely many nonzero spatial blocks. The norm normalization, geometric tail, subcritical connection and energy-enclosure budgets are explicit. The script `checks/verify_infinite_class.py` checks these rational implications and representative finite prefixes, including bonds crossing their supports. It does not decide totality of an arbitrary input program, execute arbitrary-precision critical-state oracles, or discover a new global separator. Its report and negative controls accompany S1. Universal termination for arbitrary XY configurations remains open.

Source format. This companion uses the SIADS-permitted fallback: 10-point type, 6-inch by 8-inch text area, and mathematical line numbering. The manuscript source bundle does not redistribute an isolated SIAM class file. Publisher-template installation guidance is in its README.

6 Additional separator audit, 21 October 2026

A new clean execution of the standalone separator, using its unchanged pinned scientific sources, ended with `FRESH_GLOBAL_SEPARATOR_PASS` in 180.92 seconds. The 109 local charts, 436 coordinate faces and all other boundary pieces passed, with zero unresolved cells. All 330 directly comparable scientific files were byte-identical to the archived outputs; boundary and separation records agreed after explicitly removing path, timing and derived-hash fields. All 454 fresh-output hashes were checked. This standalone run preceded the new complete execution reported in Section 7.

An independent implementation using Python integers and exact rational arithmetic verified all 715 geometric inclusions: eight Gram and inverse entries, 218 center coordinates, 109 squared transverse norms and 380 neighbor distances, as well as three scalar maxima. It computes inner products through the full-lattice degree-four Laplacian of zero extensions and checks boundary zeros. It does not import the supplied C++ interval engine. These checks validate the fixed geometry; they do not constitute an independent interval evaluation of the critical-state residuals or chart energies. Code, negative controls and complete records are included in `verification_notes/separator_2026_10_21`.

The separator's historical minimum checker prints bare decimal scalars. Its downstream consumer already adds or subtracts an exact 10^{-18} padding to every numerical baseline field it uses. For the fixed magnitudes, this dominates the decimal formatting error. This resolves the previously identified serialization concern for this interface, subject to the explicitly supported 64-bit significand environment. The separate generic nonfinite-input issue in the congruence helper must be distinguished from its reachability with these fixed inputs. No original frozen source or scientific pin was modified.

A diagnostic replay with finite/order checks completed the minimum, all 109 charts and the geometry without a guard failure. Its scientific outputs and coefficient files were byte-identical to the untouched fresh run. Exact hexadecimal sidecars were compared with the exported decimals using rational arithmetic: 8,284 chart interval records, 436 one-sided chart scalars, 715 geometry interval records, three geometry summaries and the four consumed minimum scalars passed their directional tests. This addresses the fixed separator execution and its actual exports; it neither repairs the original helper's generic NaN contract nor constitutes a second interval implementation. At that stage the connection archive was unavailable; Section 7 resolves that input obligation and reports the complete execution.

7 Complete reproduction and connection audit, 21 October 2026

The five supplied connection volumes were reconstructed into the original 441,730,856-byte archive with the exact S3 hash above. Its 129 manifested members and 44 scientific-distribution members matched. The reconstruction script pins all five volume hashes and verifies the complete archive before atomic replacement. An initially verified local archive later failed its ZIP check; it was rejected and reconstructed again before the successful payload audit. The execution note records this operational incident without assigning an unverified cause.

The unchanged three-package supervisor was rerun in a new output directory, without cached-result options or archived results as premises. It ended with `FRESH_OPTIMAL_ANNIHILATION_BARRIER_PASS` in 737.332688331604 seconds. Its certificate SHA-256 is

```
b1cbb7206a0c7492ad6990213a48480c9d25e04a5a21fe1805f0195a871f5999
```

All 548 global output hashes were checked, as well as the 454 separator and 35 connection hashes in the nested records. All mathematical fields agreed with the preceding fresh execution. The comparison also found 129 byte-identical scientific outputs, including all 109 atlas polynomials; both boundary assemblies agreed after explicit metadata normalization. The environment and compiler flags are unchanged from Section 3. No new process-tree memory peak was measured in this run.

The actual path-array audit passed, including manifest bindings, finiteness, zero boundaries, support and rational norm bounds giving $M = 6007$. Independent rational postprocessors reproduced the connection implications; seven deliberate changes to coverage, endpoint, direction or energy were rejected. The robustness checker was rebound to the new global certificate and passed with unchanged mathematical constants.

Appendix D of the article independently derives the finite load's exact zero mean and both exterior residual terms. It uses endpoint cancellation of the compact changed flux and oddness of the pure reference force, without imposing symmetry on the correction. The opposed nonlinear bond terms are combined before taking absolute values. This explains the implemented Neumann comparison and exterior formula, including the halo.

Separate instrumented copies of the connection and extension sources check finite ordered interval endpoints and export exact binary sidecars. The extension's 11 jobs passed and reproduced all 11 scientific JSON outputs and five coefficient files byte for byte; 888 export records passed exact rational directional checks. The exact positive margin between `LEVEL` and the padded saddle upper bound is approximately $1.2192816430\,10^{-21}$; rounded display values are not used to decide that comparison. An independently coded integer/rational calculation also verifies eight normal/adjoint coordinate inclusions and the identities $Gc = n$ and $c^T Gc = 1$.

The connection backend report supplies the corresponding execution and consumer-by-consumer scalar checks, including the binary constants actually compiled into dependent modules. Diagnostic copies are separate from the unchanged original sources. Their protection of this fixed execution does not repair the original congruence helper's generic NaN contract and is not a second interval engine. Historical node/segment dumps are approximate records, not numerical premises consumed by the current chain.

All compact sources, logs, exact sidecars, rational checks and binding records are in `verification_notes/connection_2026_10_21`; large raw arrays and compiled binaries remain in the separately indexed originals. This closes the previously missing-input obligation. The limits on heteroclinic, stochastic, universal and perturbed-mechanism claims in Section 5 are unchanged.